



# Tomas Bata University

## Faculty of Applied Informatics

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***Subject Name: Continuous Control (AUART/AE4SR)***

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***Topic Name: Seminar work***

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So, my number is:  $a_2= 2$ ,  $a_1=5$ ,  $a_0= 1$ ,  $b_1 = 2$ ,  $b_0 = 1$

**Task1: Find the corresponding transfer function.**

Our general equation is:

$$a_2 \cdot y''(t) + a_1 \cdot y'(t) + a_0 \cdot y(t) = b_1 \cdot u'(t) + b_0 u(t)$$

So, with my number the equation will be:

$$2y'' + 5y' + 1y = 2u' + 1u$$

Using Laplace transform with zero initial conditions:

$$\Rightarrow L\{2y''\} + L\{5y'\} + L\{1 \cdot y\} = L\{2u'\} + L\{1 \cdot u\}$$

$$\Rightarrow 2 \cdot L\{y''\} + 5 \cdot L\{y'\} + 1 \cdot L\{y\} = 2 \cdot L\{u'\} + 1 \cdot L\{u\}$$

$$\begin{aligned} \Rightarrow 2 \cdot [s^2 Y(s) - s \cdot y(0) - y'(0)] + 5 \cdot [sY(s) - y(0)] + 1 \cdot Y(s) \\ = 2 \cdot [sU(s) - u(0)] + 1 \cdot U(s) \end{aligned}$$

$$\Rightarrow 2s^2 Y(s) + 5sY(s) + 1Y(s) = 2s \cdot U(s) + 1 \cdot U(s)$$

$$\Rightarrow Y(s)[2s^2 + 5s + 1] = U(s)[2s + 1]$$

$$\Rightarrow G(s) = \frac{Y(s)}{U(s)}$$

$$G(s) = \frac{2s+1}{2s^2+5s+1} \quad [\text{this is the transfer function}]$$

**Task 2: Calculate poles, zeros and the static gain, and decide about the stability and periodicity.**

For calculating Zero: we have to set that numerator will be zero.

So, Numerator:  $N(s) = 2s + 1$

set  $N(s) = 0$

$$\Rightarrow 2s+1 = 0$$

$$\Rightarrow S = \frac{-1}{2} = -0.5$$

**One finite zero:  $z_1 = -0.5$**

Relative order:  $n - m = 2 - 1 = 1 \rightarrow$  **One zero at infinity**

For calculating Poles: we have to solve our Denominator:  $2s^2 + 5s + 1 = 0$

$$\text{So, } \Rightarrow 2s^2 + 5s + 1 = 0$$

We know, Quadratic formula

$$s = \frac{[-5 \pm \sqrt{(25 - 8)}]}{4} = \frac{[-5 \pm \sqrt{17}]}{4}$$

So, now  $s_1$  and  $s_2$

$$S_1 = \frac{-5+4.12}{4} = -0.2192 \quad S_2 = \frac{-5-4.12}{4} = -2.2808$$

**Poles:  $p_1 = -0.2192$ ,  $p_2 = -2.2808$**

**Static gain:**

For the static gain  $K = G(0)$

$$\text{So, } G(0) = \frac{(2.0+10)}{(2.0^2+5.0+1)} = \frac{1}{1} = 1$$

#Stability: Here all poles have negative real parts, therefore the system is asymptotically stable.

#Periodicity: Both poles are real (no imaginary part), therefore: System is aperiodic (non-oscillatory)

#Minimum-phase characteristic: Zero at  $s = -0.5$  (negative real part), System is minimum-phase (all zeros in left half-plane)

### TASK 3: Calculate the impulse response and draw its plot

Impulse response:  $h(t) = L^{-1}\{G(s)\}$

$$\Rightarrow G(s) = \frac{2s + 1}{2s^2 + 5s + 1}$$

$$\Rightarrow G(s) = \frac{2s + 1}{2[(s - p_1)(s - p_2)]} \quad ; p_1 = \frac{[-5 + \sqrt{17}]}{4} \text{ and } p_2 = \frac{[-5 - \sqrt{17}]}{4}$$

#### Partial fraction decomposition:

$$\Rightarrow \frac{2s + 1}{2s^2 + 5s + 1} = \frac{A}{(s - p_1)} + \frac{B}{(s - p_2)}$$

$$\Rightarrow 2s + 1 = 2A(s - p_2) + 2B(s - p_1)$$

$$\Rightarrow 2s + 1 = 2(A + B).s + 2(Ap_2 + Bp_1)$$

By Comparing coefficients:

S-term:

$$\Rightarrow 2 = 2(A + B)$$

$$\Rightarrow A + B = 1 \text{ --- Equation NO: 1}$$

constant:

$$\Rightarrow 1 = 2(Ap_2 + Bp_1)$$

$$\Rightarrow Ap_2 + Bp_1 = \frac{1}{2} \text{ --- Equation NO: 2}$$

From Equation no:1

$$\Rightarrow B = 1 - A \text{-----Equation 3}$$

Substitute In two:

$$\Rightarrow Ap_2 + (1 - A)p_1 = -\frac{1}{2}$$

$$\Rightarrow Ap_2 + p_1 - Ap_1 = -\frac{1}{2}$$

$$\Rightarrow A(p_2 - p_1) + p_1 = -\frac{1}{2}$$

Now,

$$\Rightarrow p_2 - p_1 = \left[ \frac{-5 - \sqrt{17}}{4} - \frac{-5 + \sqrt{17}}{4} \right]$$

$$\Rightarrow p_2 - p_1 = \frac{-5 - \sqrt{17} + 5 - \sqrt{17}}{4} = \frac{-2\sqrt{17}}{4} = \frac{-\sqrt{17}}{2}$$

$$\Rightarrow A\left(\frac{-\sqrt{17}}{2}\right) + p_1 = -\frac{1}{2}$$

$$\Rightarrow A\left(\frac{-\sqrt{17}}{2}\right) = -\frac{1}{2} - p_1$$

$$\Rightarrow A\left(\frac{-\sqrt{17}}{2}\right) = \frac{-(1 + 2p_1)}{2}$$

$$\Rightarrow A = \frac{-(1 + 2p_1)}{2} \times \frac{-2}{\sqrt{17}}$$

$$\Rightarrow A = \frac{1 + 2p_1}{\sqrt{17}} \text{-----Equation no: 4}$$

So,

$$\Rightarrow p_1 = \frac{[-5 + \sqrt{17}]}{4}$$

$$\Rightarrow 2p1 = \frac{[-5 + \sqrt{17}]}{2}$$

$$\Rightarrow 1 + 2p1 = 1 + \frac{[-5 + \sqrt{17}]}{2}$$

$$\Rightarrow 1 + 2p1 = \frac{2 - 5 + \sqrt{17}}{2}$$

$$\Rightarrow 1 + 2p1 = \frac{-3 + \sqrt{17}}{2}$$

Now, in equation 4:

$$\Rightarrow A = \frac{-\frac{3 + \sqrt{17}}{2}}{\sqrt{17}}$$

$$\Rightarrow A = -\frac{3 + \sqrt{17}}{2\sqrt{17}} \text{----- VALUE OF A.}$$

And B will be:

$$\Rightarrow B = 1 - A \Rightarrow B = 1 - \left(-\frac{3 + \sqrt{17}}{2\sqrt{17}}\right)$$

$$\Rightarrow B = \frac{2\sqrt{17} + 3 - \sqrt{17}}{2\sqrt{17}}$$

$$\Rightarrow A + B = -\frac{3 + \sqrt{17}}{2\sqrt{17}} + \frac{2\sqrt{17} + 3 - \sqrt{17}}{2\sqrt{17}}$$

$$\Rightarrow A + B = \frac{-3 + \sqrt{17} + 2\sqrt{17} + 3 - \sqrt{17}}{2\sqrt{17}} = \frac{2\sqrt{17}}{2\sqrt{17}} = 1$$

So, now

$$\Rightarrow G(s) = \frac{\left(-\frac{3 + \sqrt{17}}{2\sqrt{17}}\right) \cdot 1}{(s - p1)} + \frac{\left(\frac{3 + \sqrt{17}}{2\sqrt{17}}\right) \cdot 1}{(s - p2)}$$

### Impulse response by using Inverse Laplace Transform:

$$\Rightarrow h(t) = \left[ \frac{-3 + \sqrt{17}}{2\sqrt{17}} \right] \cdot e^{(p_1 \cdot t)} + \left[ \frac{3 + \sqrt{17}}{2\sqrt{17}} \right] \cdot e^{p_2 \cdot t} \text{ --- for } t \geq 0$$

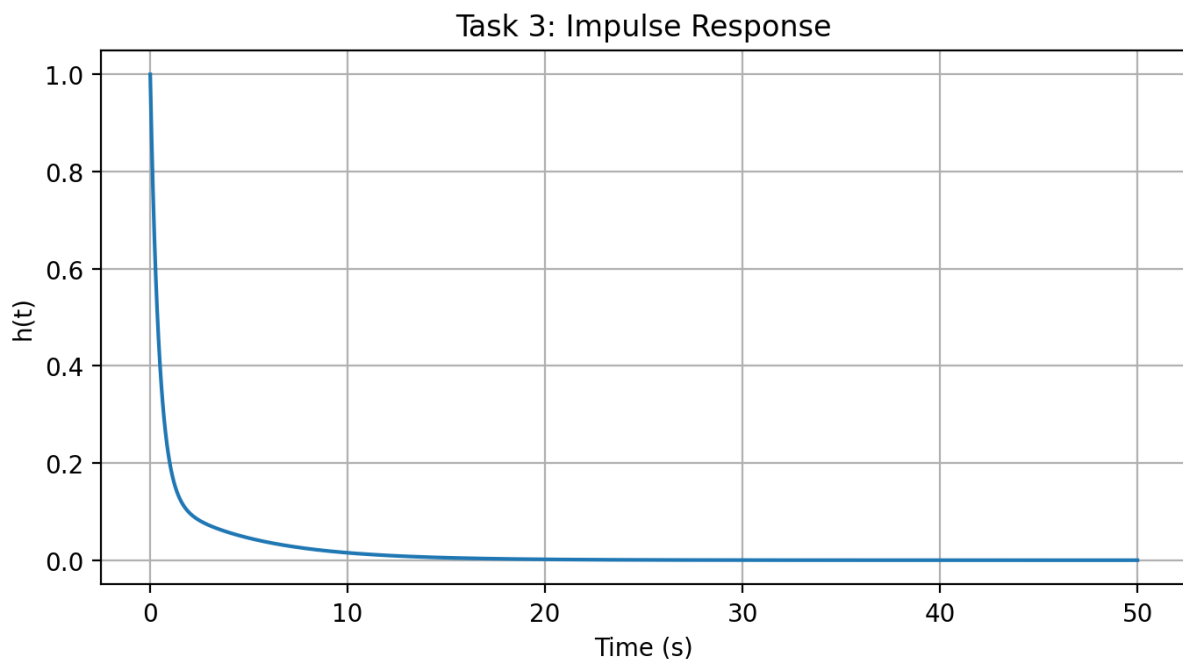
So,

$$\Rightarrow p_1 = \frac{[-5 + \sqrt{17}]}{4} = -0.2192 \text{ and } p_2 = \frac{[-5 - \sqrt{17}]}{4} = -2.2808$$

$$\Rightarrow A = -\frac{3 + \sqrt{17}}{2\sqrt{17}} = 0.1362 \text{ and } B = \frac{2\sqrt{17} + 3 - \sqrt{17}}{2\sqrt{17}} = 0.8638$$

Finally,

$$\Rightarrow h(t) = 0.1362 \cdot e^{-0.2192t} + 0.8638 \cdot e^{-2.2808t}$$



**Task4: Calculate the step response and draw its plot by using a suitable SW.**

Ans:

$$\Rightarrow G(s) = \frac{2s + 1}{2s^2 + 5s + 1}$$

$$\Rightarrow u(t) = 1(t)$$

$$\Rightarrow U(s) = s1$$

$$\Rightarrow Y(s) = G(s) \cdot U(s)$$

$$\Rightarrow Y(s) = \frac{2s + 1}{2s^2 + 5s + 1} \cdot \frac{1}{s}$$

$$\Rightarrow Y(s) = \frac{2s + 1}{s(2s^2 + 5s + 1)}$$

*Factor Denominator:*

$$\Rightarrow 2s^2 + 5s + 1 = 2(s + 0.219)(s + 2.281)$$

$$\Rightarrow Y(s) = \frac{2s + 1}{2s(s + 0.219)(s + 2.281)}$$

*Partttial Fraction:*

$$\Rightarrow Y(s) = \frac{A}{s} + \frac{B}{(s - 0.219)} + \frac{C}{(s + 2.281)}$$

*Multiply both sides:*

$$\Rightarrow \frac{2s + 1}{2} = A(s + 0.219)(s + 2.281) + Bs(s + 2.281) + Cs(s + 0.219)$$

*Solve for A, B, C*

$$\Rightarrow s = 0$$

$$\Rightarrow \frac{1}{2} = A(0.219)(2.281)$$

$$\Rightarrow A = 1 - - - - - one$$

$$\Rightarrow s = -0.219$$

$$\Rightarrow \frac{2(-0.219) + 1}{2} = B(-0.219)(2.062)$$

$$\Rightarrow 0.281 = B(-0.451)$$

$$\Rightarrow B = -0.623 \text{ --- --- --- --- --- } -Two$$

$$\Rightarrow s = -2.281$$

$$\Rightarrow \frac{2(-2.281) + 1}{2} = C(-2.281)(-2.062)$$

$$\Rightarrow -1.781 = C(4.704)$$

$$\Rightarrow C = -0.378 \text{ --- --- --- --- --- } -Three$$

*Laplace Expression*

$$\Rightarrow Y(s) = \frac{1}{s} - \frac{0.623}{s + 0.219} - \frac{0.378}{s + 2.281}$$

*Inverse Laplace Transform*

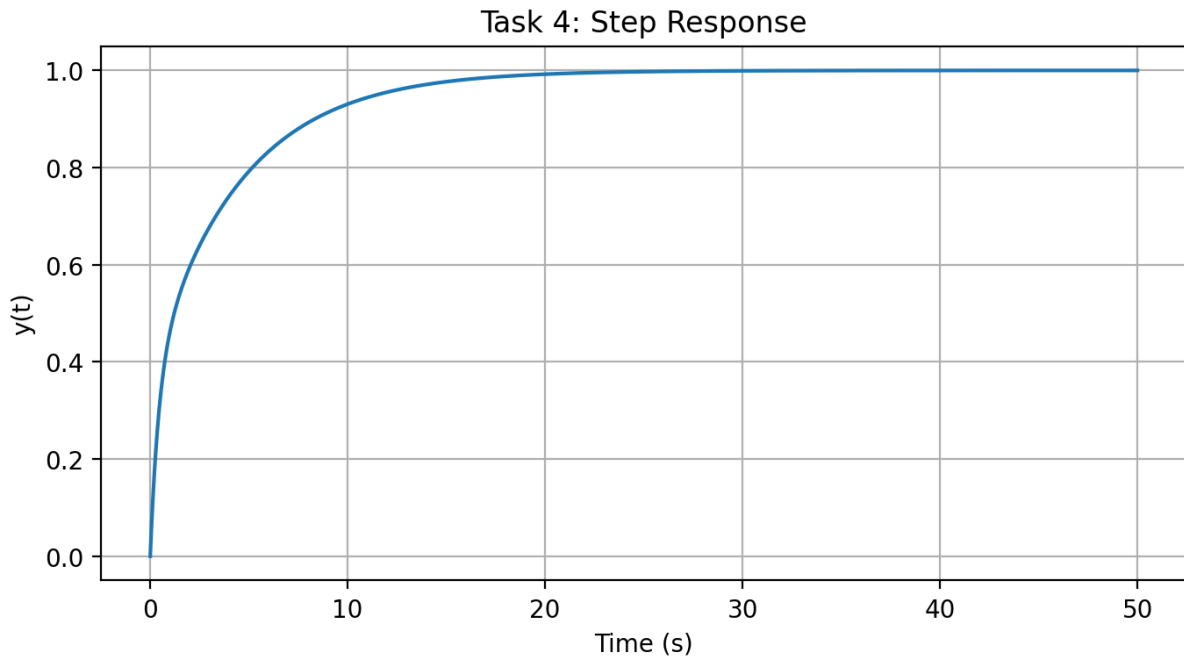
$$\Rightarrow L^{-1}\left(\frac{1}{s}\right) = 1$$

$$\Rightarrow L^{-1}\left(\frac{1}{s + a}\right) = e^{-at}$$

$$\Rightarrow y(t) = 1 - 0.623e^{-0.219t} - 0.378e^{-2.281t} \text{ --- --- --- --- ---}$$

*--- Step Response*

So, from the equation we can say the step response is stable and there is no oscillation.



**Task5: Calculate the corresponding frequency transfer function expressed in both the algebraic (real&imaginary component) form and goniometric (exponential) form.**

$$\Rightarrow G(s) = \frac{2s + 1}{2s^2 + 5s + 1}$$

Here,  $\Rightarrow s = j\omega$

$$\Rightarrow G(j\omega) = \frac{2(j\omega) + 1}{2(j\omega)^2 + 5(j\omega) + 1}$$

*simplify by numerator and denominator*

*Numerator:*

$$\Rightarrow 2(j\omega) + 1 = 1 + j2\omega$$

*Denominator :*

$$\Rightarrow j\omega^2 = -\omega^2 \Rightarrow 2(j\omega)^2 = -2\omega^2$$

*so,*

$$\Rightarrow 2j\omega^2 + 5(j\omega) + 1 = (1 - 2\omega^2) + j5\omega$$

*Final frequency tranfer function is:*

$$\Rightarrow G(j\omega) = \frac{1 + j2\omega}{(1 - 2\omega^2) + j5\omega} \text{----- Final one.}$$

*Algebraic Form:*

$$\Rightarrow G(j\omega) = \frac{a + jb}{c + jd} \text{----- } [a = 1, b = 2\omega, c = 1 - 2\omega^2, d = 5\omega]$$

$$\Rightarrow G(j\omega) = \frac{(1 + j2\omega)(c - jb)}{c^2 + d^2}$$

*Expand Numerator*

$$\Rightarrow (1 + j2\omega)(c - j5\omega)$$

$$\text{Multiply real part} \Rightarrow c + 10\omega^2$$

$$\text{Multiply imaginary part} \Rightarrow -5\omega + 2\omega c$$

*Substitute  $c = 1 - 2\omega^2$  in 'c' position*

$$\Rightarrow (1 - 2\omega^2) + 10\omega^2 = 1 + 8\omega^2 \text{----- Real part}$$

*Then*

$$\Rightarrow -5\omega + 2\omega(1 - 2\omega^2)$$

$$\Rightarrow -5\omega + 2\omega - 4\omega^3$$

$$\Rightarrow -3\omega - 4\omega^3 \text{----- Imaginary part}$$

*Denominator*

$$\Rightarrow (1 - 2\omega^2)^2 + (5\omega)^2$$

$$\Rightarrow (1 - 4\omega^2 + 4\omega^4) + 25\omega^2 = \\ > 1 + 21\omega^2 + 4\omega^4 \text{----- Denominator}$$

*Now,*

$$\Rightarrow G(j\omega) = \frac{1 + 8\omega^2}{1 + 21\omega^2 + 4\omega^4} + j \frac{-3\omega - 4\omega^3}{1 + 21\omega^2 + 4\omega^4} \text{-----} \\ \text{-- Algebraic form.}$$

*Goniometric Form*

$$\Rightarrow G(j\omega) = |G(j\omega)| e^{j\varphi(\omega)}$$

*Magnitude:*

$$\Rightarrow |G(j\omega)| = \frac{\sqrt{1^2 + (2\omega)^2}}{\sqrt{(1 + 21\omega^2 + 4\omega^4)}}$$

$$\Rightarrow |G(j\omega)| = \frac{\sqrt{1^2 + 4\omega^2}}{\sqrt{(1 + 21\omega^2 + 4\omega^4)}} \text{----- Magnitude}$$

*Phase:*

$$\varphi(\omega) = \arg(1 + j2\omega) - \arg((1 - 2\omega^2) + j5\omega)$$

$$\Rightarrow \varphi(\omega) = \tan^{-1}(2\omega) - \tan^{-1}\left(\frac{5\omega}{1 - 2\omega^2}\right) \text{----- Phase}$$

**Q6. Choose at least 10 frequency values and insert them into the algebraic form of the frequency transfer function (i.e., calculate the function values). Display the results in a table and also as the Nyquist plot.**

Ans: By using algebraic form

$$\Rightarrow G(j\omega) = \frac{1 + 8\omega^2}{1 + 21\omega^2 + 4\omega^4} + j \frac{-3\omega - 4\omega^3}{1 + 21\omega^2 + 4\omega^4}$$

$\Rightarrow$  for  $\omega$  I have selected 10 values such as:

$$\omega = 0, 0.5, 1, 2, 3, 5, 10, 20, 50, 100$$

Now,

$$\Rightarrow \operatorname{Re}(G) = \frac{1 + 8\omega^2}{1 + 21\omega^2 + 4\omega^4}$$

$$\Rightarrow \operatorname{Im}(G) = \frac{-3\omega - 4\omega^3}{1 + 21\omega^2 + 4\omega^4}$$

Table for algebraic value:

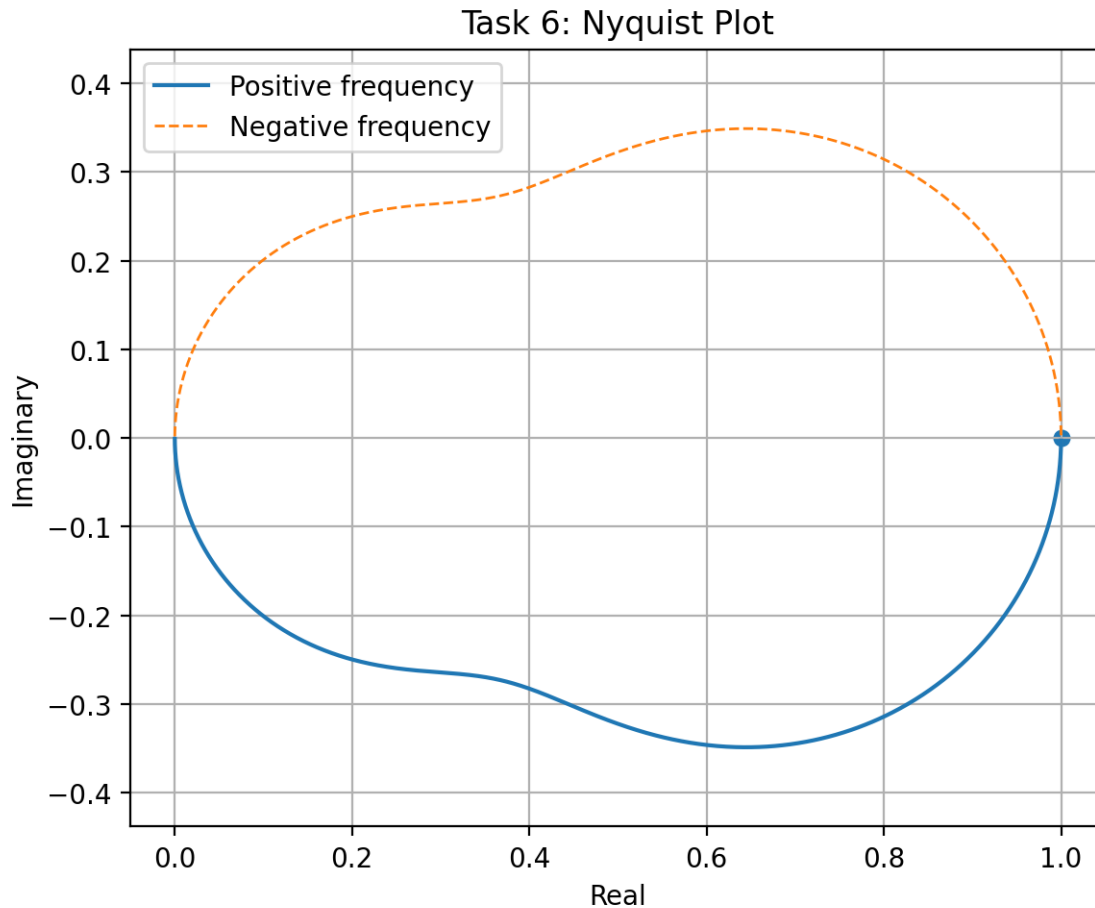
| $\omega(\frac{red}{s})$ | Re(G(j $\omega$ )) | Im(G(j $\omega$ )) |
|-------------------------|--------------------|--------------------|
| 0                       | 1.0000             | 0.0000             |
| 0.5                     | 0.4615             | -0.3077            |
| 1                       | 0.3462             | -0.2692            |
| 2                       | 0.2215             | -0.2550            |
| 3                       | 0.1420             | -0.2276            |
| 5                       | 0.0664             | -0.1702            |
| 10                      | 0.0190             | -0.0957            |
| 20                      | 0.0049             | -0.0494            |
| 50                      | 0.0008             | -0.0200            |

Interpretation:

When omega is zero:  $\Rightarrow G(j0) = 1 + j0$

And  $\omega = \infty$  then  $G(j\omega) = 0$

So, when imaginary part is negative then the curve will be downward.



**Q7. Choose at least 10 frequency values and insert them into the goniometric form of the frequency transfer function (i.e., calculate the function values). Display the results in a table and also as Bode plots.**

Ans: Goniometric form:

$$\Rightarrow |G(j\omega)| = \frac{\sqrt{1^2 + 4\omega^2}}{\sqrt{(1 + 21\omega^2 + 4\omega^4)}}$$

$$\Rightarrow \varphi(\omega) = \tan^{-1}(2\omega) - \tan^{-1}\left(\frac{5\omega}{1 - 2\omega^2}\right)$$

Now, I have convert to the Magnitude to dB.

$$\Rightarrow |G(j\omega)|_{dB} = 20 \log_{10} |G(j\omega)|$$

then  $\omega = 0.5, 1, 2, 3, 5, 10, 20, 50, 100$

| $\omega \left( \frac{\text{rad}}{s} \right)$ | $ G(j\omega) $ | Magnitude (dB) | Phase (deg) |
|----------------------------------------------|----------------|----------------|-------------|
|----------------------------------------------|----------------|----------------|-------------|

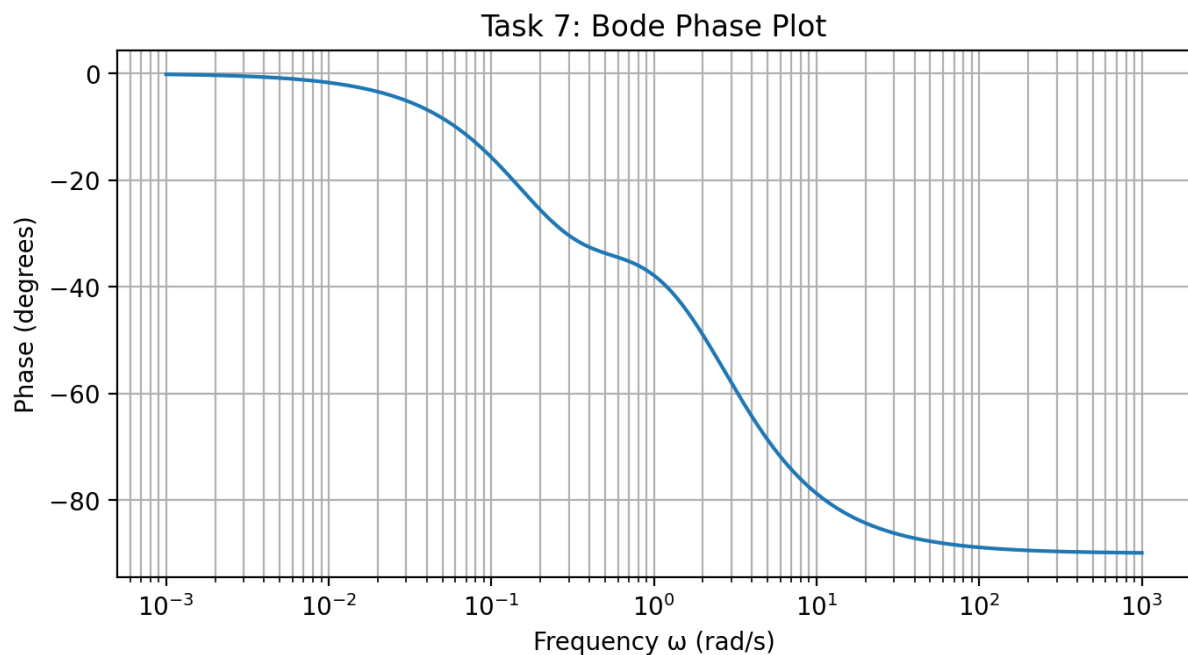
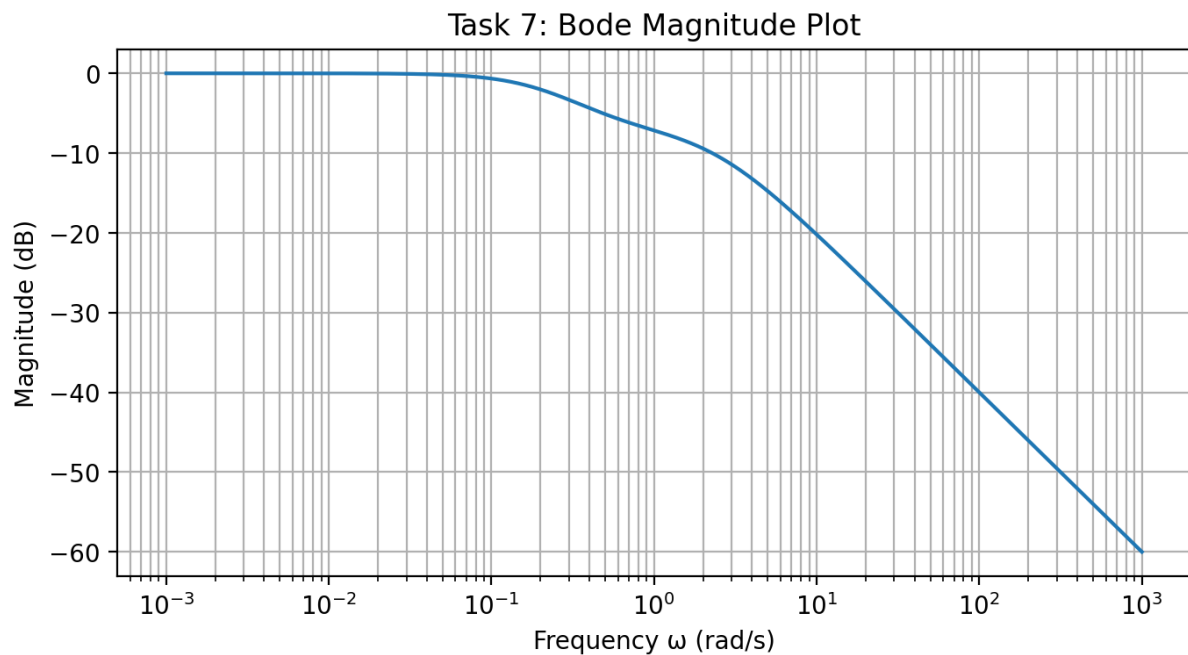
|     |        |           |         |
|-----|--------|-----------|---------|
| 0.5 | 0.5547 | -5.12dB   | -33.69° |
| 1   | 0.4385 | -7.16 dB  | -37.87° |
| 2   | 0.3378 | -9.43 dB  | -49.03° |
| 3   | 0.2682 | -11.43 dB | -58.05° |
| 5   | 0.1827 | -14.76 dB | -68.70° |
| 10  | 0.0975 | -20.22 dB | -78.75° |
| 20  | 0.0496 | -26.09 dB | -84.35° |
| 50  | 0.0200 | -33.98 dB | -87.71° |
| 100 | 0.0100 | -40.00 dB | -88.85° |

### **Magnitude:**

1. It starts at nearly 0 dB
2. Decreases as frequency increases
3. Shows low-pass system behavior

### **Phase:**

1. Starts near 0°
2. Decreases gradually toward -90°



**Q8. Let the simple negative feedback control system (1DoF) be with a step-wise reference and a step-wise load disturbance signals. By using the polynomial algebraic method, design a controller satisfying control feedback stability, (asymptotic) reference tracking, and disturbance rejection. Let feedback poles be set as  $-a_0, -a_1$  and  $-0.5(a_0 + a_1)$ . Display control responses  $u(t)$ ,  $e(t)$ ,  $y(t)$  (by using Matlab).**

Ans:  $\Rightarrow G(s) = \frac{2s+1}{2s^2+5s+1}$

Here,

$$\Rightarrow A(s) = 2s^2 + 5s + 1$$

$$\Rightarrow B(s) = 2s + 1$$

$$\text{Polynomial controller: } C(s) = \frac{R(s)}{S(s)} = \frac{r_1s + r_0}{s}$$

$$\text{So, } R(s) = r_1s + r_0 \text{ and } S(s) = s.$$

*Closed – Loop characteristic*

*Ours desired poles:  $-a_0, -a_1$  and  $-0.5(a_0 + a_1)$ ;*

$$\text{so } -a_0 = -1, -a_1 = -5 \text{ and } -0.5(a_0 + a_1) = -3$$

*So, closed loop poles are :  $-1, -5$  and  $-3$  ;*

$\Rightarrow$  Polynomial characteristic is:

$$\Rightarrow (s + 1)(s + 5)(s + 3);$$

$$\Rightarrow (s^2 + 6s + 5)(s + 3);$$

$$\Rightarrow s^3 + 9s^2 + 23s + 15;$$

*Characteristic equation is:*

$$\Rightarrow A(s)S(s) + B(s)R(s);$$

$$\Rightarrow (2s^2 + 5s + 1)(s) + \{(2s + 1)(r_1s + r_0);$$

$$\Rightarrow 2s^3 + 5s^2 + s + 2r_1s^2 + (2r_0 + r_1)s + r_0;$$

$$\Rightarrow 2s^3 + (5 + 2r_1)s^2 + (1 + 2r_0 + r_1)s + r_0; \text{ --- Equation}$$

$\Rightarrow$  Normalization

$$\Rightarrow s^3 + \frac{5 + 2r_1}{2}s^2 + \frac{1 + 2r_0 + r_1}{2}s + \frac{r_0}{2}$$

*so now for  $s^2$ ,*

$$\Rightarrow \frac{5 + 2r_1}{2} = 9;$$

$$\Rightarrow 5 + 2r_1 = 18; \Rightarrow r_1 = 6.5$$

$$\text{For } s \Rightarrow \frac{1 + 2r_0 + r_1}{2} = 23;$$

$$\Rightarrow 1 + 2r_0 + r_1 = 46$$

$$\Rightarrow 2r_0 = 46 - 1 - 6.5 = 38.5;$$

$$\Rightarrow r_0 = 19.25$$

*for constant*

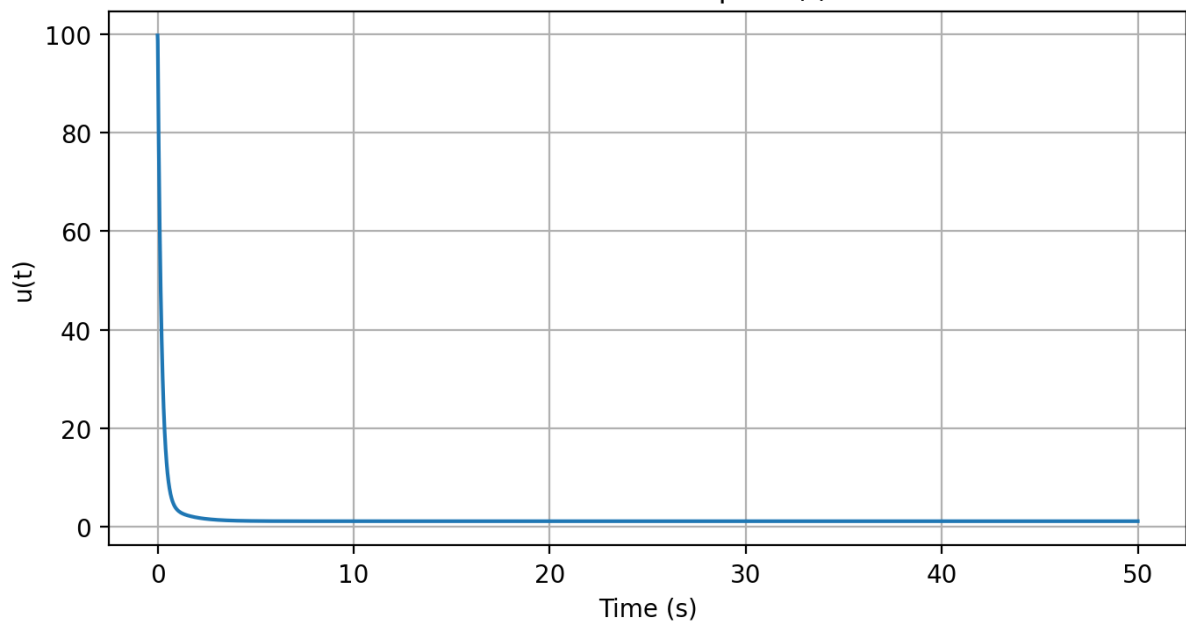
$$\Rightarrow \frac{r_0}{2} = 15;$$

$$\Rightarrow r_0 = 30;$$

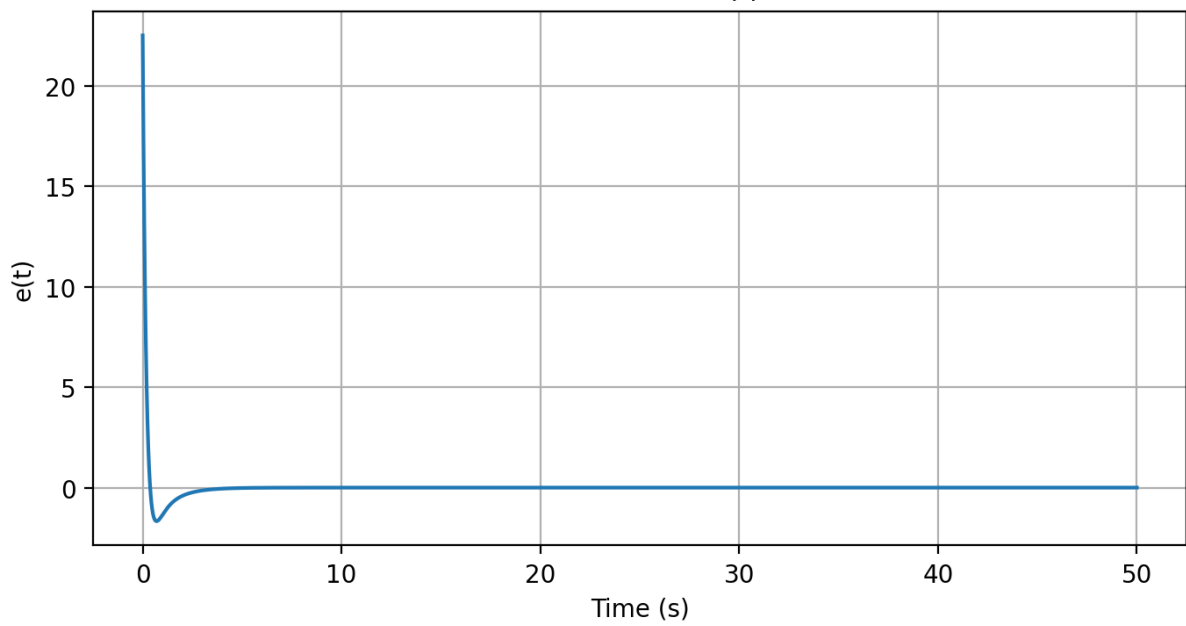
*Final controller*

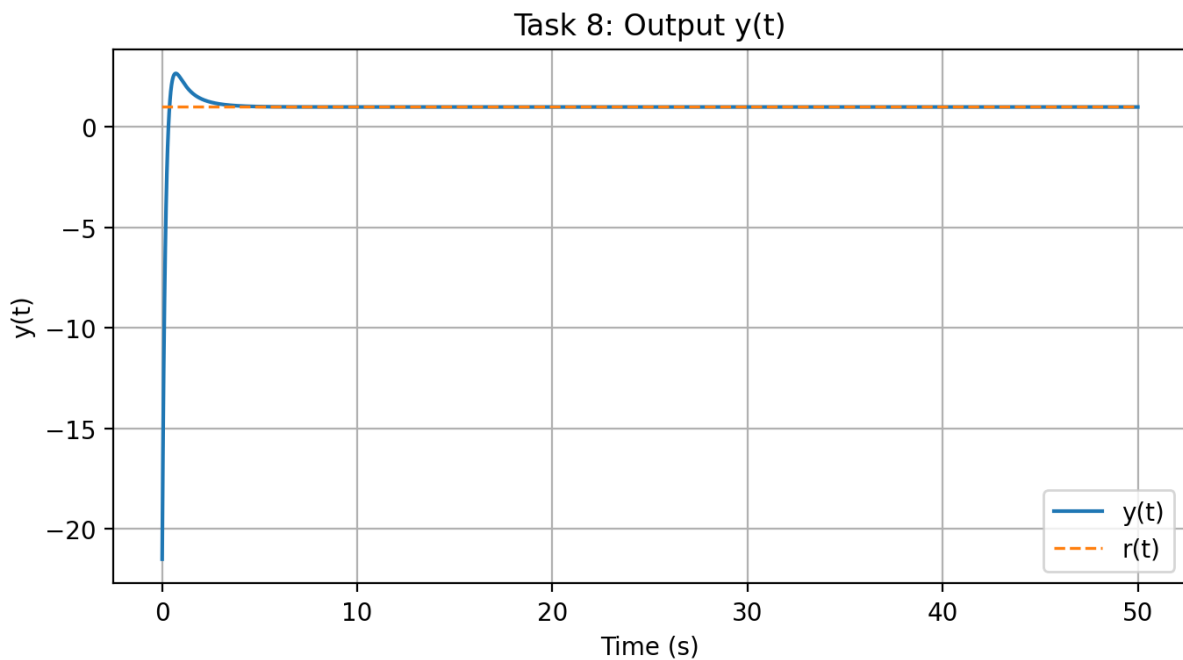
$$\Rightarrow C(s) = \frac{6.5s + 19.25}{s}$$

Task 8: Control Input  $u(t)$



Task 8: Error  $e(t)$





**Q9. Design the PI(D) controller via another (arbitrary) simple control design method (e.g., using the model step response characteristics). Display control responses  $u(t)$ ,  $e(t)$ ,  $y(t)$  (by using Matlab).**

Ans: My system is  $\Rightarrow G(s) = \frac{2s+1}{2s^2+5s+1}$

*where PI Controller:*

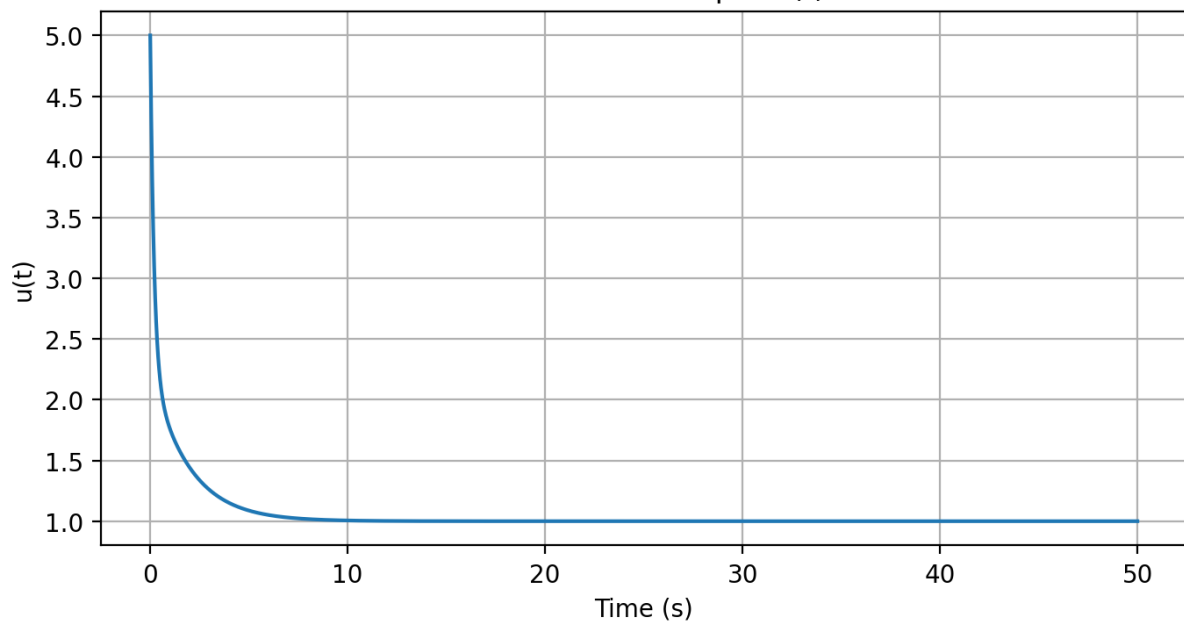
$$\Rightarrow C(s) = K_p + \frac{K_i}{s};$$

*Now,  $K_p = 5 = \text{faster response}$ .*

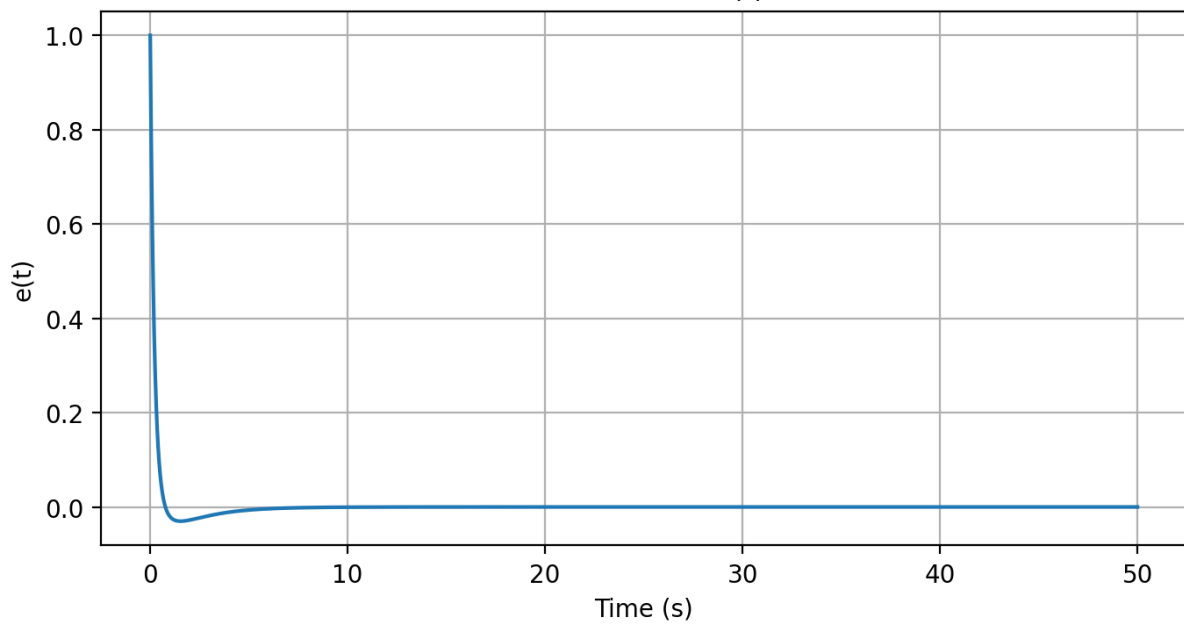
*and  $K_i = 10 = \text{eliminate steady error}$ .*

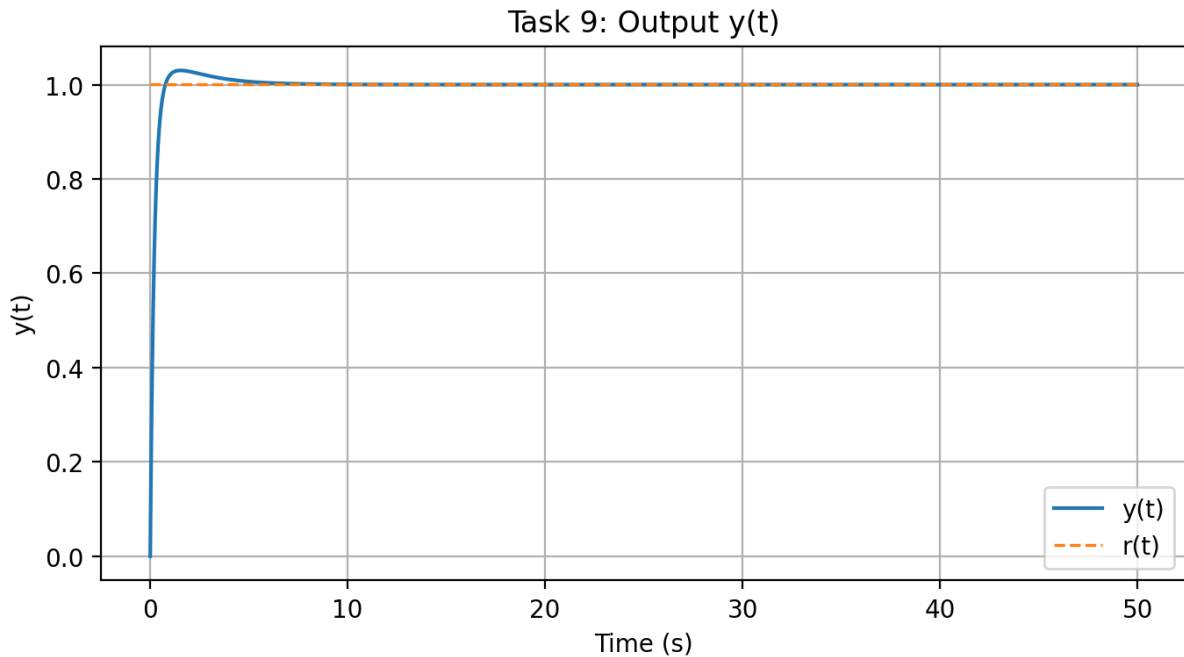
$$\text{so, } C(s) = 5 + \frac{10}{s}$$

Task 9: Control Input  $u(t)$



Task 9: Error  $e(t)$





**Q10. Verify control feedback asymptotic (Lyapunov) stability for control systems from tasks #8 and #9 by using an arbitrary stability criterion.**

$$\text{Ans: } \Rightarrow C(s) = 5 + \frac{10}{s} = \frac{5s+10}{s}$$

*Close – loop characteristic equation:*

$$\Rightarrow 1 + C(s)G(s) = 0;$$

$$\Rightarrow 1 + \frac{(5s + 10)(2s + 1)}{s(2s^2 + 5s + 1)} = 0;$$

$$\Rightarrow s(2s^2 + 5s + 1) + (5s + 10)(2s + 1) = 0;$$

*First term,*  $\Rightarrow 2s^3 + 5s^2 + s;$

*Second term,*

$$\Rightarrow (5s + 10)(2s + 1) = 10s^2 + 25s + 10;$$

*Total*  $\Rightarrow 2s^3 + 15s^2 + 26s + 10 = 0;$

roots ([2 15 26 10])

The poles are approximately:

$$s = -5.39, s = -2.05, s = -0.05$$

All poles have negative real parts, so the Task 9 control system is also:

Asymptotically stable

### **Final Statement**

Both control systems from Task 8 and Task 9 are asymptotically stable because all closed-loop poles lie in the left half-plane.

**Q11. Find two different state-space descriptions of the system model.**

$$\text{Ans: } \Rightarrow G(s) = \frac{2s+1}{2s^2+5s+1};$$

*Divide by 2:*

$$\Rightarrow G(s) = \frac{s + 0.5}{s^2 + 2.5s + 0.5};$$

*Controllable Canonical Form:*

$$\Rightarrow x' = Ax + Bu$$

$$\Rightarrow y = Cx + Du;$$

$$A1 = \begin{bmatrix} 0 & 1 \\ -0.5 & -2.5 \end{bmatrix}, B1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, C1 = [0.5 \quad 1],$$

$$D1 = 0$$

*Now,*

$$\Rightarrow x' = \begin{bmatrix} 0 & 1 \\ -0.5 & -2.5 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$\Rightarrow y = [0.5 \quad 1]x$$

*Observable Canonical Form:*

$$A1 = \begin{bmatrix} -2.5 & 1 \\ -0.5 & 0 \end{bmatrix}, B1 = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}, C1 = [1 \quad 0], D1 = 0$$

$$\Rightarrow x' = \begin{bmatrix} -2.5 & 1 \\ -0.5 & 0 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0.5 \end{bmatrix} u$$

$$\Rightarrow y = [1 \quad 0]x$$

Both state-space models represent the same transfer function.

**Q12. For both the state-space descriptions, perform (reversely) the input-output differential equation (or transfer function) calculation.**

Ans: For this task I need,  $\Rightarrow G(s) = C(s.I - A)^{-1}B + D$

*From task11 model 1:*

$$A1 = \begin{bmatrix} 0 & 1 \\ -0.5 & -2.5 \end{bmatrix}, B1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, C1 = [0.5 \quad 1],$$

$$D1 = 0$$

*Now,*

$$\Rightarrow s.I - A1 = \begin{bmatrix} s & 1 \\ -0.5 & s + 2.5 \end{bmatrix}$$

$$\Rightarrow \det(s.I - A1) = s(s + 2.5) + 0.5$$

$$\Rightarrow 2s^2 + 5s + 1$$

And then,  $\Rightarrow G(s) = \frac{s + 0.5}{s^2 + 2.5s + 0.5};$

$\Rightarrow G(s) = \frac{2s + 1}{2s^2 + 5s + 1};$  [multiplying by 2]

Model2:

$A2 = \begin{bmatrix} -2.5 & 1 \\ -0.5 & 0 \end{bmatrix}, B2 = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}, C2 = [1 \quad 0], D2 = 0$

Now,  $\Rightarrow s.I - A1 = \begin{bmatrix} s + 2.5 & -1 \\ 0.5 & s \end{bmatrix}$

$\Rightarrow \det(s.I - A1) = s(s + 2.5) + 0.5$

$\Rightarrow s^2 + 2.5s + 0.5$

$\Rightarrow G(s) = \frac{2s + 1}{2s^2 + 5s + 1};$  [multiplying by 2]

**Q13. Verify that both the state-space descriptions are controllable and observable.**

Ans: If a system is controllable the:

$$\text{rank}(W_c) = n;$$

If a system is controllable the:

$$\text{rank}(W_o) = n;$$

And, the system order is:  $n=2$

**State-Space model1:**

$$A1 = \begin{bmatrix} 0 & 1 \\ -0.5 & -2.5 \end{bmatrix}, B1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, C1 = [0.5 \quad 1],$$

$$D1 = 0$$

*Controllability Matrix:*

$$\Rightarrow W_c = [B1 \quad A1B1];$$

$$\Rightarrow W_c = \begin{bmatrix} 0 & 1 \\ -0.5 & -2.5 \end{bmatrix}$$

$$\text{And, } \text{rank}(W_c) = 2;$$

***so the model is controllable.***

**Observability Matrix:**

$$\Rightarrow W_o = \begin{bmatrix} C1 \\ C1A1 \end{bmatrix}$$

$$\Rightarrow C1A1 = [-0.5 \quad -2]$$

$$\Rightarrow W_o = \begin{bmatrix} 0.5 & 1 \\ -0.5 & -2 \end{bmatrix}$$

$$\Rightarrow \text{rank}(W_o) = 2;$$

**Model 1 is observable.**

**State-Space model2:**

$$A2 = \begin{bmatrix} -2.5 & 1 \\ -0.5 & 0 \end{bmatrix}, B2 = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}, C2 = [1 \quad 0], D2 = 0$$

**Controllability Matrix:**

$$\Rightarrow W_c = [B \quad A^2B];$$

$$\Rightarrow A^2B = \begin{bmatrix} -2 \\ -0.5 \end{bmatrix}$$

$$\Rightarrow W_c = \begin{bmatrix} 1 & -2 \\ 0.5 & -0.5 \end{bmatrix}$$

$$\Rightarrow \text{rank}(W_c) = 2$$

**Model 2 is controllable.**

Now,

Observability Matrix:

$$\Rightarrow W_o = \begin{bmatrix} C \\ C^2A \end{bmatrix}$$

$$\Rightarrow C^2A = [-2.5 \quad 1]$$

$$\Rightarrow W_o = \begin{bmatrix} 1 & 0 \\ -2.5 & 1 \end{bmatrix}$$

$$\Rightarrow \text{Rank}(W_o) = 2$$

**Model 2 is observable.**

Both state-space models are **controllable and observable** because all controllability and observability matrices have rank 2, which is equal to the system order.

**Q14. Calculate the standard fundamental matrix for one (selected) state-space description from task #11.**

Ans: For a state-space model  $x' = Ax + Bu$ ,  $\Phi(s)$

$$\Rightarrow \Phi(s) = (s.I - A)^{-1}$$

Next,

$$\Rightarrow sI - A1 = \begin{bmatrix} s & -1 \\ 0.5 & s + 2.5 \end{bmatrix}$$

The determinant of  $sI - A1$ :

$$\Rightarrow \det(sI - A1) = s(s + 2.5) + 0.5$$

$$\Rightarrow s^2 + 2.5s + 0.5$$

The inverse of  $sI - A1$  is:

$$\Rightarrow (sI - A1)^{-1} = \frac{1}{\det(sI - A1)} \begin{bmatrix} s + 2.5 & 1 \\ -0.5 & s \end{bmatrix}$$

$$\Rightarrow \Phi(s) = \frac{1}{s^2 + 2.5s + 0.5} \begin{bmatrix} s + 2.5 & 1 \\ -0.5 & s \end{bmatrix}$$

The matrix  $\Phi(s)$  is obtained by calculating  $(sI - A)^{-1}$ , which is the inverse of the matrix  $sI - A_1$ , and scaling it by the determinant of  $sI - A_1$ . This matrix represents the state evolution of the system for zero initial conditions.

**Q15. Solve the state-space equation for zero initial conditions and  $u(t) = 1$  (by means of the standard fundamental matrix). Then calculate the output and compare the result with that from task #4.**

Ans: Task 15 asks to solve the state-space equation for zero initial conditions and  $u(t) = 1(t)$

The state-space equation:

$$x' = Ax(t) + Bu(t)$$

With zero initial conditions:

$$X(0) = 0;$$

$$\text{Output: } y(t) = Cx(t) + Du(t)$$

State-Space Model 1:

$$A1 = \begin{bmatrix} 0 & 1 \\ -0.5 & -2.5 \end{bmatrix}, B1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, C1 = [0.5 \quad 1],$$

$$D1 = 0$$

State equation:

$$\begin{bmatrix} x'1 \\ x'2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -0.5 & -2.5 \end{bmatrix} \begin{bmatrix} x1 \\ x2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

$$\text{so, } x'1 = x'2$$

$$\Rightarrow x'2 = -0.5x1 - 2.5x2 + u(t)$$

$$\Rightarrow x'2 = -0.5x1 - 2.5x2 + 1$$

By using laplace transform:

$$\Rightarrow X2(s) = sX1(s);$$

$$\Rightarrow sX2(s) = -0.5X1(s) - 2.5X2(s) + \frac{1}{s}$$

Substitute  $X_2(s) = sX_1(s)$ :

$$\Rightarrow s^2X1(s) = -0.5X1(s) - 2.5sX1(s) + \frac{1}{s}$$

$$\Rightarrow (s^2 + 2.5s + 0.5)X1(s) = \frac{1}{s}$$

Now,

$$\Rightarrow X1(s) = \frac{1}{s(s^2 + 2.5s + 0.5)}$$

$$\text{and } \Rightarrow X2(s) = \frac{1}{(s^2 + 2.5s + 0.5)}$$

$$\text{So, } p1 = -0.2192 \text{ and } p2 = -2.2808$$

so,

$$\Rightarrow x_1(t) = 2 - 2.2127e^{-0.2192t} + 0.2127e^{-2.2808t}$$

$$\Rightarrow x_2(t) = 0.4851e^{-0.2192t} + 0.4851e^{-2.2808t}$$

*Calculate output:*

$$\Rightarrow y(t) = Cx(t) + Du(t)$$

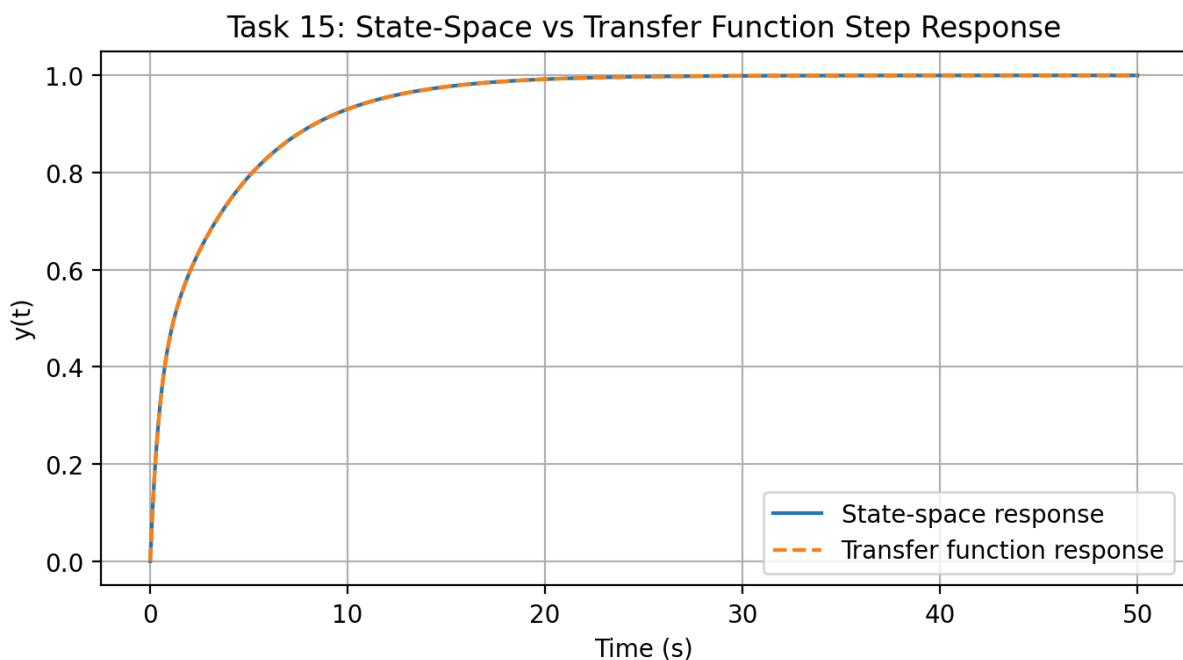
$$D = 0;$$

$$\Rightarrow y(t) = 0.5x_1(t) + x_2(t)$$

*Substitute  $x_1(t)$  and  $x_2(t)$*

$$\Rightarrow y(t) = 0.5(2 - 2.2127e^{-0.2192t} + 0.2127e^{-2.2808t}) + 0.4851e^{-0.2192t} + 0.4851e^{-2.2808t}$$

$$\Rightarrow y(t) = 1 - 0.6213e^{-0.2192t} - 0.3787e^{-2.2808t}$$



**Q16. For a suitable state-space model from task #11, design a proportional state-feedback controller satisfying control feedback stability and (asymptotic) reference tracking. Let feedback poles be set as  $-a_0$ ,  $-a_1$  and  $-0.5(a_0 + a_1)$ . Display control responses  $u(t)$ ,  $e(t)$ ,  $y(t)$  (by using Matlab).**

Ans:

$$A = \begin{bmatrix} 0 & 1 \\ -0.5 & -2.5 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, C = [0.5 \quad 1],$$

$$\text{Closed loop matrix: } A_{cl} = \begin{bmatrix} 0 & 1 & 0 \\ -0.5 - k_1 & -2.5 - k_2 & -k_i \\ -0.5 & -1 & 0 \end{bmatrix}$$

Integral state:  $x' = r - y$

The control law is:

$$\Rightarrow u = -Kx + k_i x_i - - - K = \begin{bmatrix} k_1 \\ k_2 \end{bmatrix}$$

Poles are:

$$-a_0 = -1, -a_1 = -5 \text{ and } -0.5(a_0 + a_1) = -3;$$

Characteristic polynomial:

$$\Rightarrow (s + 1)(s + 3)(s + 5);$$

$$\Rightarrow s^3 + 9s^2 + 23s + 15$$

So,

$$\Rightarrow 2.5 + k_2 = 9;$$

$$\Rightarrow k_2 = 6.5$$

$$\text{then, } \Rightarrow -0.5k_i = 15;$$

$$\Rightarrow k_i = -30;$$

$$\Rightarrow 0.5 + k_1 - k_i = 23;$$

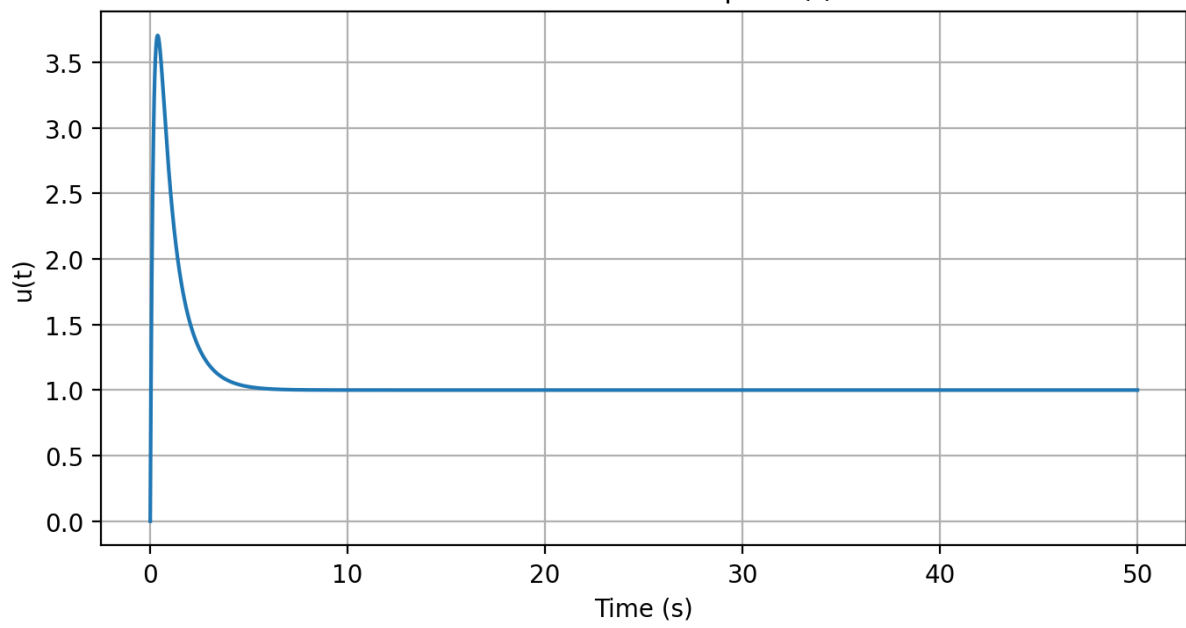
$$\Rightarrow k_1 = -7.5$$

Now,

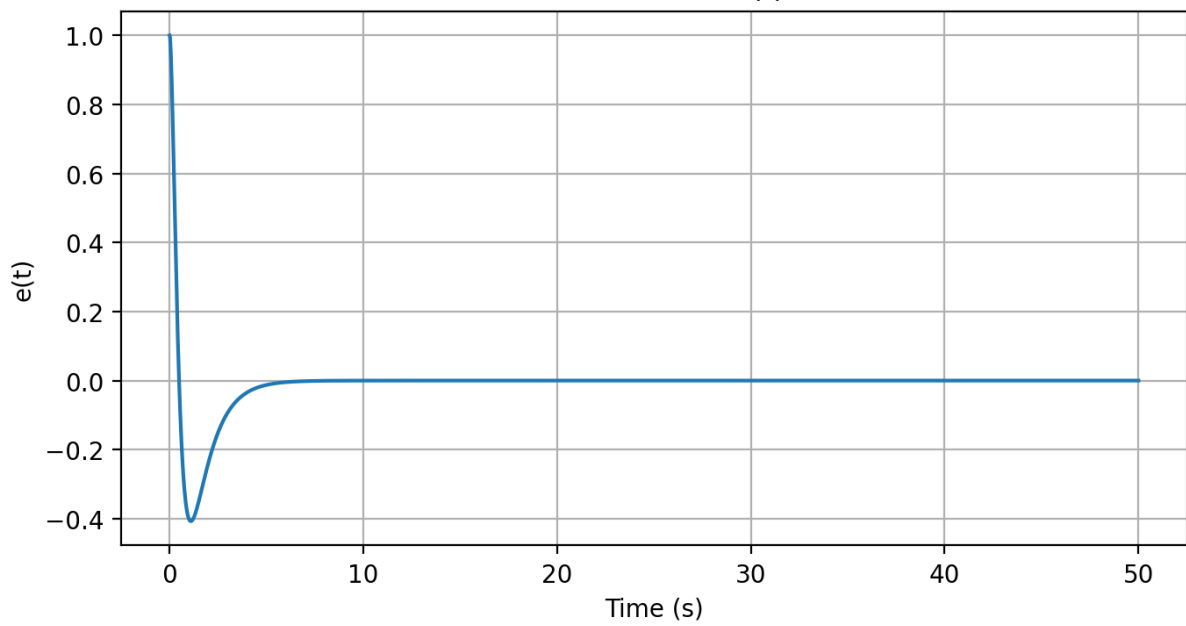
$$\Rightarrow u = -k_1 x_1 - k_2 x_2 - k_i x_i$$

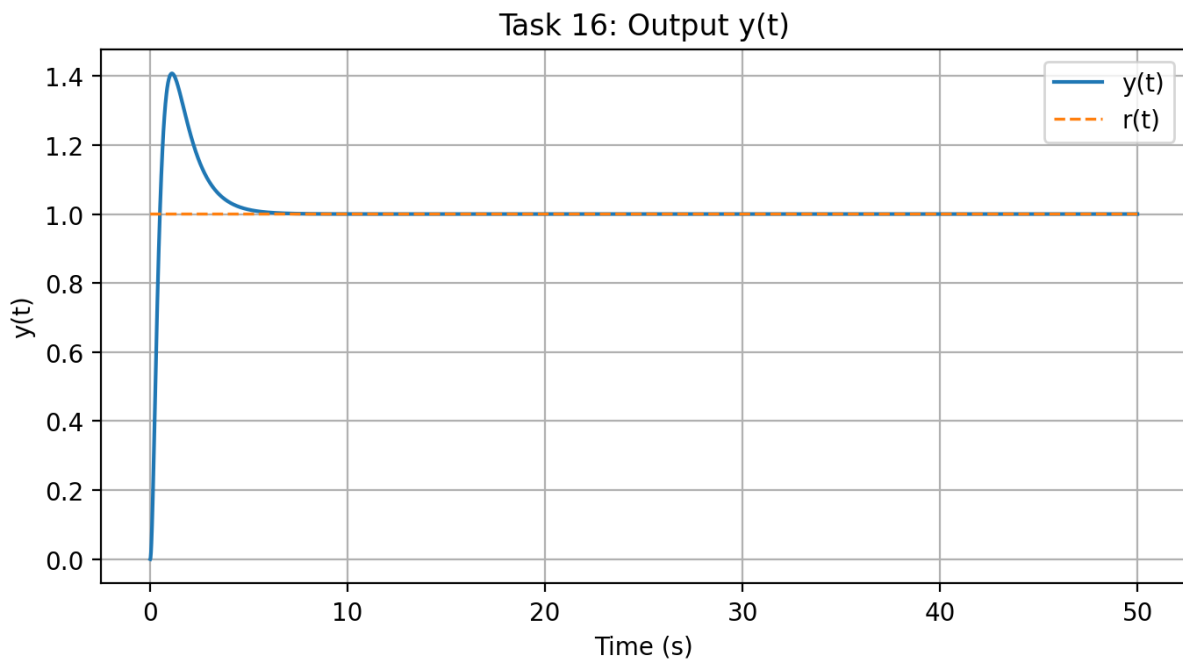
$$\Rightarrow u = 7.5x_1 - 6.5x_2 + 30x_i$$

Task 16: Control Input  $u(t)$



Task 16: Error  $e(t)$





**Q17. Compare (in words) the obtained results from tasks #8, #9, and #16.**

Ans: In **Task 8**, the controller was designed using the **polynomial algebraic method**. The closed-loop poles were placed at:

$S = -1, S = -3 \text{ AND } S = -5$

This gives a stable response with asymptotic reference tracking and disturbance rejection.

Task9:  $C(s) = 5 + \frac{10}{s}$

This controller is easier to design, and it also reduces the steady-state error to zero. However, compared with Task 8, it is less systematic because the gains were chosen by tuning instead of exact pole placement.

Task 16: A state-feedback controller with integral action was designed. The closed-loop poles were also placed at:

$S = -1, S = -3 \text{ AND } S = -5$

This method gives stable control and asymptotic reference tracking. It is more advanced because it uses the state-space model and state feedback.

[End]